Pre-FOMC Announcement Drift After 2020

Draft

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Abstract

Abstract to be written after the holdout result (task T10/T11).

1 Introduction

Lucca and Moench (2015) document large average excess equity returns in the hours before scheduled FOMC policy statements. We ask whether that premium survives after 2020 and whether it is concentrated in meetings that follow high-uncertainty states, as the uncertainty-based account of Hu et al. (2022) suggests. A premium that shrinks after publication would be consistent with the general decay of published predictors (McLean and Pontiff, 2016).

We fix the specification, state threshold and inference procedure in a pre-analysis plan, estimate on 2010–2019 intraday data, and run the 2020–2026 holdout once. In the holdout the anxious-state premium is 37.8 bps but is not statistically distinguishable from zero with year-clustered wild bootstrap inference ($p = 0.149$), and the anxious-minus-calm contrast is -2.6 bps ($p = 0.946$). Returns in meeting windows exceed those on matched-VIX non-meeting days. With seven clusters and nine calm meetings the holdout has limited power, so we report bounds rather than evidence of absence. The contribution is a disciplined out-of-sample check, not a causal estimate.

2 Related Literature

Lucca and Moench (2015) report large average excess equity returns before scheduled FOMC announcements (about 49 bps over the 24 hours ending roughly 15 minutes before the announcement, 131 scheduled meetings from September 1994 to March 2011, as stated in the New York Fed staff report version), find that similar pre-announcement returns appear in other major international equity indices but not in Treasuries or money market futures, and note that other macroeconomic announcements do not produce comparable returns. Hu et al. (2022) find large overnight returns with no abnormal variance before nonfarm payrolls, ISM and GDP releases, and propose a model in which heightened uncertainty about the impact of upcoming news, proxied by the VIX, is priced and then resolves at the announcement. Martello and Ribeiro study state dependence of the FOMC premium on implied volatility; a secondary summary of that paper reports an average drift of 109 bps when implied volatility is above its prior median and 9.7 bps below (UNVERIFIED: taken from a third-party summary, not from the paper's text). McLean and Pontiff (2016) study 97 cross-sectional predictors and report portfolio returns 26% lower out of sample and 58% lower post-publication; they study cross-sectional predictors, not announcement-window returns, so we use them only as a benchmark for expected decay. Our holdout tests the state-dependent prediction on data that none of these papers used.

3 Data

The outcome is the log return from the prior day's 16:00 price to the 13:45 price on the meeting day, net of the risk-free rate, in basis points, on intraday S&P 500 prices. The state variable is the VIX level five trading days before the meeting. The development sample covers 77 scheduled meetings in 2010–2019 and 2,239 non-meeting days. The holdout covers 2020–2026: 53 scheduled meetings, of which 52 pass the validity filter and one is dropped. Half-day sessions and days with missing trades are flagged and excluded by the panel builder, which logs row counts at every merge. The anxious/calm threshold is the development-sample median VIX, 15.07, fixed before the holdout was touched. The press-conference indicator is known only from 2021, so the 2020 meetings are excluded from that subsample (status UNVERIFIED for 2020).

4 Empirical Design

4.1 Identification

The object of interest is the return earned in the window that ends before the FOMC statement is released, in excess of the risk-free rate. Close-to-close daily returns include the announcement reaction and therefore measure a different quantity, so the primary test uses intraday prices. The comparison of interest is between meetings held in calm and in anxious states, where the state is the VIX close five trading days before the meeting (lagged one week) and the anxious threshold is fixed at the development-sample median of 15.07 VIX points. The threshold is not re-estimated on the holdout. The secondary specification replaces the indicator with the continuous lagged VIX.

Identification rests on the timing of the window and on the state being predetermined. Lagging the state by a week limits contamination from anticipation of the meeting itself. The claim is a conditional association between pre-announcement returns and the prior volatility state, not a causal effect of the Committee's actions, and the paper is framed accordingly.

4.2 Threats to inference

Episode clustering. Anxious meetings concentrate in a small number of volatility episodes, so the effective number of independent observations is far smaller than the number of meetings. We cluster by calendar year as an ex-ante proxy for episodes, use CR1 cluster-robust standard errors, and treat the wild cluster bootstrap-t as primary because the holdout has few clusters. Asymptotic t statistics are descriptive only.

Power. With a small number of independent episodes the design can detect a large premium but cannot distinguish a half-sized premium from zero. We therefore report minimum detectable effects from a clustered power simulation using realized volatility, and present non-rejections as bounds on the premium rather than as evidence of absence.

Release-time measurement. Statement release times are assigned by rule rather than observed per meeting, so the window may include a few minutes of the announcement reaction in some early meetings. This affects the level of the estimate more than the state comparison. *Update (verified against Federal Reserve release pages).* Eight development meetings (2011-04-27, 2011-06-22, 2011-11-02, 2012-01-25, 2012-04-25, 2012-06-20, 2012-09-13, 2012-12-12) released the statement at about 12:20–12:35 ET, not 14:15 ET, so their 14:00 window end follows the announcement. The holdout

(2020 onward) is unaffected: its statements are released at 14:00 ET. Appendix A.5 reports the development-sample effect of this error.

Non-FOMC confounds. A premium measured on FOMC days could reflect generic high-volatility or pre-event compensation. The matched-VIX placebo compares FOMC windows with non-FOMC windows on days with similar lagged VIX, so a Fed-specific premium requires an excess over matched days.

State measurement. VIX comparability has changed over time as short-dated option volume has grown. This matters mainly for the exploratory VIX-change test, which is outside the frozen specification.

Specification search. The specification, threshold, outcome window and inference procedure are fixed in the pre-analysis plan before any estimate on post-2019 data, and the holdout is run once. Tests outside the plan are labelled exploratory and cannot change the primary conclusion.

5 Results

All holdout estimates come from a single run of the frozen specification; secondary tests are labelled exploratory. Standard errors are clustered by year (seven clusters), so we report wild cluster bootstrap p-values alongside.

Primary holdout test. Table 1 reports the primary specification. The anxious-state premium is 37.8 bps ($t = 1.91$, wild $p = 0.149$) against 19.3 bps in development; its sign matches development but it does not reject zero at 5%. The key contrast b (anxious minus calm) is -2.6 bps (wild $p = 0.946$): it does not replicate the development difference of 26.0 bps. The reason is the calm cell, whose mean rises from -6.6 to 40.4 bps. That cell has only nine meetings, so b is weakly identified (Figures 1 and 2). The continuous VIX slope is 5.02 bps per point ($t = 2.48$), but it is a single-sample, seven-cluster estimate. The matched-VIX placebo on non-meeting days gives a Fed-minus-matched difference of 55.5 bps ($p = 0.002$) for all meetings and 38.3 bps ($p = 0.030$) for anxious meetings, so the window return around meetings exceeds that on comparable non-meeting days.

Table 1: Primary specification, development and holdout. Standard errors (year-clustered) in parentheses.

	Development	Holdout
Calm mean a (bps)	-6.6 (3.3)	40.4 (5.4)
Anxious$-$calm b (bps)	26.0 (11.6)	-2.6 (21.9)
Anxious premium $a + b$ (bps)	19.3 (11.4)	37.8 (19.8)
Wild-bootstrap $p(b = 0)$	0.059	0.946
Wild-bootstrap $p(a + b = 0)$	—	0.149
VIX slope c (bps/pt)	1.60 (0.97)	5.02 (2.02)
Placebo (all, $n = 52$): Fed$-$matched (bps)	—	55.5 ($p = 0.002$)
Placebo (anxious, $n = 43$): Fed$-$matched (bps)	—	38.3 ($p = 0.030$)
Placebo (calm, $n = 9$): Fed$-$matched (bps)	—	35.2 ($p = 0.209$)

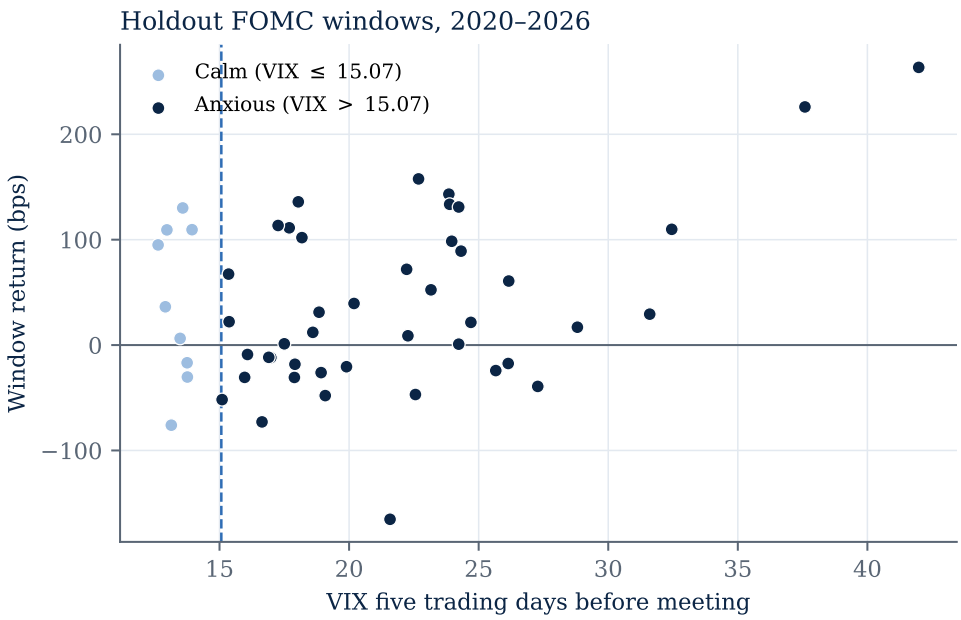


Figure 1: Holdout meeting-window returns against lagged VIX.

Exploratory secondary tests. The VIX-change test finds no effect: the slope on the five-day VIX change is -3.39 ($t = -0.47$, wild $p = 0.876$). The press-conference subsample (45 meetings from 2021 on; the Federal Reserve lists a press conference at every 2020 meeting as well, so the subsample excludes 2020 only because that flag was unverified when the test was run once) gives an anxious premium of 28.3 bps (wild $p = 0.221$). No meeting in the sample has the press-conference indicator equal to zero, so the comparison with non-press meetings cannot be made. Table 2 shows the anxious premium across thresholds, leave-one-year-out samples, winsorizing and no risk-free adjustment. It ranges from 26.6 to 69.5 bps. No wild p-value falls below 0.05, and the smallest is 0.092.

Interpretation. With seven clusters and nine calm meetings, the holdout is under-powered for the state contrast: the minimum detectable effect for the anxious half is 41.9 bps at the development volatility. The evidence supports a positive meeting-window return, especially against matched non-meeting days. It does not confirm the development finding that the premium is concentrated in anxious states.

6 Discussion

The holdout supports a positive meeting-window return, particularly relative to matched non-meeting days, but does not replicate the development finding that the premium is concentrated in anxious states. Three readings are consistent with the data. First, the premium may be roughly state-independent after 2020; the calm cell mean rose from -6.6 to 40.4 bps. Second, the state contrast may be present but undetectable: the calm cell has nine meetings and the minimum detectable effect is 41.9 bps. Third, the 2020–2022 episode, in which VIX was high and the communication regime changed, may dominate a seven-cluster sample. The data cannot separate these.

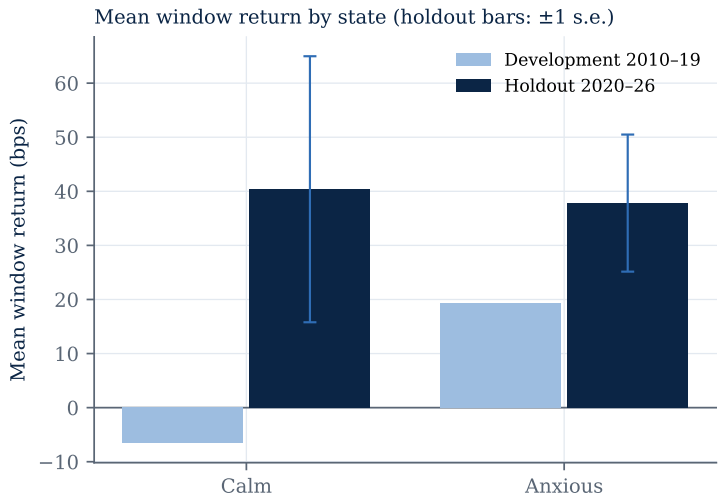


Figure 2: Mean window return by state, development versus holdout.

Limitations are as stated in the design: release times are assigned by rule, and for eight 2011–2012 development meetings the rule is wrong (the statement came at about 12:30 ET), which inflates the development benchmark (Appendix
refapp:sens), clustering by year is a proxy for volatility episodes, and the VIX-change and press-conference tests are exploratory. The comparison with the post-publication decline in McLean and Pontiff (2016) is loose: the holdout premium does not fall, but an announcement-window return is not a cross-sectional anomaly portfolio, and the sample is too small to measure a decline precisely.

7 Conclusion

A single pre-specified holdout run finds a positive pre-announcement return in 2020–2026 that exceeds matched non-meeting days, but it does not confirm that the premium is concentrated in high-uncertainty meetings. Power is limited by few independent episodes and a thin calm cell. Longer samples or per-meeting release times are the natural next steps.

References

Grace Xing Hu, Jun Pan, Jiang Wang, and Haoxiang Zhu. Premium for heightened uncertainty: Explaining pre-announcement market returns. *Journal of Financial Economics*, 145(3):909–936, 2022.

David O. Lucca and Emanuel Moench. The pre-FOMC announcement drift. *Journal of Finance*, 70(1):329–371, 2015.

Vitor Martello and Ruy Ribeiro. Pre-FOMC announcement relief. SSRN working paper 3286745, November 2018.

R. David McLean and Jeffrey Pontiff. Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32, 2016.

Table 2: Robustness of the holdout anxious premium (exploratory).

Specification	n	n_{anx}	Prem. (bps)	s.e.	Wild p
press=1 anxious premium	45	37	28.3	20.5	0.221
2021+ (flag-known sample)	45	37	28.3	20.5	0.221
baseline thr 15.07	52	43	37.8	19.8	0.149
threshold 18	52	30	51.6	20.7	0.092
threshold 20	52	23	59.2	22.3	0.194
threshold 25	52	9	69.5	30.4	0.51
holdout-median threshold	52	26	48.7	25.2	0.284
drop 2020	45	37	28.3	20.5	0.221
drop 2021	44	35	51.9	17.2	0.114
drop 2022	44	35	26.6	20.4	0.256
drop 2023	44	39	41.6	21.7	0.199
drop 2024	44	39	34.8	21.9	0.251
drop 2025	44	35	37.7	24.7	0.3
drop 2026	47	38	43.8	21.6	0.111
excl 2020 (covid)	45	37	28.3	20.5	0.221
winsorized 5/95	52	43	35.9	16.2	0.121
no rf adjustment	52	43	38.8	19.8	0.135

A Notation and Additional Material

A.1 Notation

All returns are in basis points unless stated. Meeting m indexes scheduled FOMC meetings; $t(m)$ is the meeting date and $t(m)^-$ the previous valid session.

Symbol	Definition	Unit
P_{prev}	16:00 ET last price on the previous valid session	index points
P_{end}	Last price at the end of the pre-announcement window: 13:45 ET (14:00 ET release, from 2013-03-01 and throughout the holdout) or 14:00 ET (14:15 ET release, earlier)	index points
rf_m	Risk-free rate over the same interval (daily file)	bps
r_m	Outcome: $10^4 \left[\ln(P_{\text{end}}/P_{\text{prev}})\right] - rf_m$	bps
VIX_m	VIX close five trading days before the meeting date	VIX points
τ	Anxiety threshold, fixed at 15.07 (development-sample median)	VIX points
A_m	$\mathbf{1}\{\text{VIX}_m > \tau\}$, the anxious indicator	0/1
a	Intercept: premium in calm meetings	bps
b	Anxious minus calm difference	bps
$a + b$	Premium in anxious meetings (primary test statistic)	bps
c	Slope of r_m on VIX_m in the secondary specification	bps per point
ε_m	Disturbance	bps
$g(m)$	Cluster (calendar year, a provisional proxy for volatility episodes)	—
G	Number of clusters (about seven in the holdout)	—

Table 3: Variables and parameters.

A.2 Specifications

Primary:

$$r_m = a + b\,A_m + \varepsilon_m, \qquad H_1 : a + b > 0. \tag{1}$$

Secondary, reported alongside and not substituted:

$$r_m = a + c\,\mathrm{VIX}_m + \varepsilon_m. \tag{2}$$

Inference uses CR1 cluster-robust standard errors by $g(m)$, wild cluster bootstrap-t (999 draws, seed 1) for $b = 0$ and $a + b = 0$, and a matched-VIX permutation placebo against non-FOMC days (2000 draws, seed 2). With few clusters the bootstrap p-values are primary and asymptotic t statistics descriptive.

A.3 Hypotheses

H1 $a + b > 0$ in the holdout.

H2 The holdout anxious premium is below the development estimate of 19.3 bps.

H3 $b > 0$.

A.4 Sample definitions

Development sample: scheduled meetings from 2010-06 to 2019-12-31. Holdout: scheduled meetings from 2020-01-01. Unscheduled meetings are excluded. A meeting is usable if the meeting day and previous session are valid (no half-day, missing-trade or cash-closed flag) and r_m and VIX_m are non-missing. Release times are a calendar rule, not observed per meeting (UNVERIFIED).

A.5 Development-sample sensitivity

Table values are anxious-state premia in basis points on the 77 development meetings (threshold 15.07), computed by `code/11_sensitivity_dev.py`; the holdout is not involved. Wild-bootstrap p-values are for the anxious-minus-calm contrast.

Variant	n	Anxious	Contrast	Wild p
Baseline (14:00 / 13:45 end)	77	19.3	26.0	0.059
End 13:45, all meetings	77	20.2	26.8	0.062
End 13:30, all meetings	77	19.2	26.5	0.081
End 12:00, all meetings	77	19.8	22.5	0.070
Drop 8 midday-release meetings	69	7.5	14.1	0.127
End 12:00 for the 8, else baseline	77	12.6	19.2	0.057

The eight midday-release meetings raise the development anxious premium from 7.5 to 19.3 bps, so the development benchmark against which the holdout was compared is partly inflated by announcement reactions. The sign of the state contrast survives in every variant. The frozen primary specification and the holdout result are unchanged.

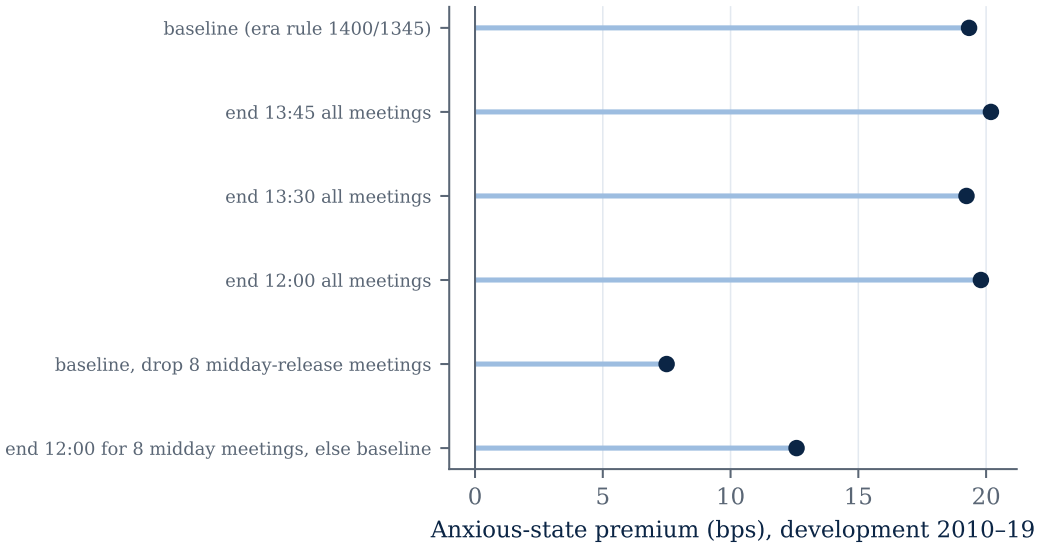


Figure 3: Development anxious-state premium under alternative window ends.